

Project Design Phase

Proposed Solution Template

Date	29 September 2026
Team ID	SWTID-2026-9139
Project Name	AI WeatherWise
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Users may need to check weather information across locations and interpret raw weather data before planning travel or outdoor activities. They need convenient, centralized and actionable weather information.
2.	Idea / Solution description	AI WeatherWise is a RESTful backend built with Node.js and Express.js, using MongoDB and Mongoose for data persistence. It provides user registration and login, current weather retrieval, favorite-location management and weather-based recommendations through API endpoints. JWT authentication, password hashing, input sanitization and fallback handling are included.
3.	Novelty / Uniqueness	The solution combines weather retrieval with personalized weather-based guidance in a modular backend. Instead of only returning weather metrics, the recommendation service is designed to provide useful activity and clothing guidance, while fallback support helps the API continue functioning when an external AI service is unavailable.
4.	Social Impact / Customer Satisfaction	The system can help travellers and outdoor-activity users make better-informed plans by providing weather information and practical recommendations in one service. Favorite locations and secure authentication improve convenience for repeated use.
5.	Business Model (Revenue Model)	The current project is an academic prototype and does not define a direct revenue model. A future deployment could support a free basic service with optional premium API access or advanced personalized weather insights.
6.	Scalability of the Solution	The modular Node.js/Express architecture can be extended with additional weather data, more recommendation features, notification services, additional locations and a dedicated frontend. MongoDB/Mongoose provides structured persistence for users and favorite locations.